Physics 468/668: Statistical Physics **Spring 2019**

Lecture 4: The Statistical Basis of Thermodynamics

Lecturer: Qiang Zhu *Scribes: scribe-name1,2,3*

4.1 The entropy of mixing and Gibbs paradox

Recall that we finally derived the expression for the entropy of ideal gas in the previous lecture, with a couple of assumptions.

1. noninteracting particles
2. distinguishable

$$S(N,V,E) = Nk\ln\left[\frac{V}{h^3}\left(\frac{4\pi mE}{3N}\right)^{3/2}\right] + \frac{3}{2}Nk \tag{4.1}$$

If we really dig into the equation, we readily observe that the entropy is not an extensive property of the system! That is, if we increase the size of the system by a factor α, keeping the intensive variables unchanged, then the entropy of the system, which should also increase by the same factor, does not do so. This is due to the presence of $\ln V$ term. The conclusion here simply means that the entropy of this system is different from the sum of the entropies of its parts. This is known as *Gibbs paradox*.

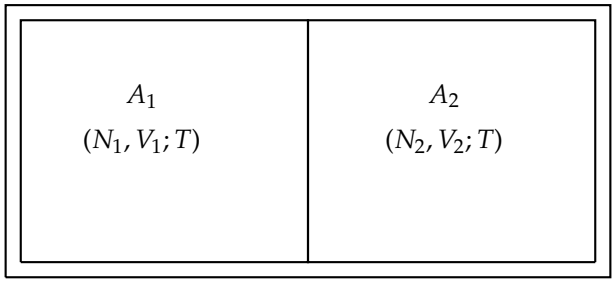


Figure 4.1: A schematic of two physical systems in thermal contact.

Gibbs visualized the mixing of two gases 1 and 2 with the same initial temperature T. Clearly, the temperature of the mixture would also be the same. For each gas, we compute the entropy as follows,

$$S_i = N_i k\ln V_i + \frac{2}{3}N_i k\left[1 + \ln(\frac{2\pi m_i kT}{h^2})\right] \tag{4.2}$$

After mixing, the total is,

$$S_T = \sum_{i=1}^{2}\left[N_i k\ln V + \frac{2}{3}N_i k\left[1 + \ln(\frac{2\pi m_i kT}{h^2})\right]\right] \tag{4.3}$$

where $V = V_1 + V_2$. Thus the net increase in the value of S called entropy of mixing is

$$\Delta S = S_T - \sum_{i=1}^{2} S_i = k\left[N_1 \ln \frac{V_1 + V_2}{V_1} + N_2 \ln \frac{V_1 + V_2}{V_2}\right] \tag{4.4}$$

The quantity ΔS is indeed positive according to 2nd thermodynamic law. For this case when two gases have same density, the equation becomes,

$$\Delta S^* = k\left[N_1 \ln \frac{N_1 + N_2}{N_1} + N_2 \ln \frac{N_1 + N_2}{N_2}\right] \tag{4.5}$$

Now, if we further assume that gas 1 and gas 2 has same N, V, T. By this equation, it seems that the system will gain some entropy after mixing. But we know this is not physical!

Gibbs realized that one can avoid this issue by simply adding one more term in the original expression of S. This is due to that we slip the multiplicity function to account for the fact that molecules in a gas are *indistinguishable*, and the factor is exactly $N!$.

$$\begin{aligned} S(N,V,E) &= Nk\ln\left[\frac{V}{h^3}\left(\frac{4\pi mE}{3N}\right)^{3/2}\right] + \frac{3}{2}Nk - k\ln N! \\ &= Nk\ln\left[\frac{V}{h^3}\left(\frac{4\pi mE}{3N}\right)^{3/2}\right] + \frac{3}{2}Nk - k(N\ln N - N) \\ &= Nk\ln\left[\frac{V}{Nh^3}\left(\frac{4\pi mE}{3N}\right)^{3/2}\right] + \frac{5}{2}Nk \end{aligned} \tag{4.6}$$

Note here we used the Stirling approximation

$$\ln N! = N \ln N - N \tag{4.7}$$

In this case,

$$\Delta S^* = k\left[(N_1 + N_2)\ln\frac{V_1 + V_2}{N_1 + N_2} + N_1 \ln \frac{V_1}{N_1} + N_2 \ln \frac{V_2}{N_2}\right] \tag{4.8}$$

Therefore, the paradox could be resolved by counting. The equation is generally referred as the *Sackur − Tetrode* equation.

Let us see some consequences due to Gibbs's revision. First of all, we compute the energy,

$$E(S,V,N) = \frac{3h^2 N}{4\pi m V^{2/3}} \exp\left(\frac{2S}{3Nk} - 1\right) \qquad \text{previous} \tag{4.9}$$

$$E(S,V,N) = \frac{3h^2 N^{5/3}}{4\pi m V^{2/3}} \exp\left(\frac{2S}{3Nk} - \frac{5}{3}\right) \qquad \text{new} \tag{4.10}$$

This will make E as a truly extensive quantity. The most important consequence is on the chemical potential,

$$\mu = \left(\frac{\partial E}{\partial N}\right)_{V,S} = E\left[\frac{5}{3N} - \frac{2S}{3N^2 k}\right] \tag{4.11}$$

Applying the ideal gas law, we obtain

$$\mu = \frac{1}{N}[E + PV - TS] = \frac{G}{N} \tag{4.12}$$

Where G is the Gibbs free energy of the system. Express μ as a function of (N, V, T)

$$\mu(N, V, T) = kT\ln\left[\frac{N}{V}\left(\frac{h^2}{2\pi mkT}\right)^{3/2}\right] \tag{4.13}$$

We can also obtain F by $G - PV$

At room temperature and atmospheric pressure, $V/N = 4.2 \times 10^{26} m^3$, $(h^2/2\pi mkT)^{2/3}$ for helium is $1.3 \times 10^{-31} m^3$ Therefore, the logarithm is 12.7. Therefore, the chemical potential is -0.32 eV (helium) and -0.42 eV (for argon), respectively.

Chemical potential is usually negative!

4.2 How to count the microstates

In the previous section, we saw that the entropy of an N-particle system has to be divided by the factor of $k\ln(N!)$, which implies an *ad hoc* reduction of the number of microstates. It is natural to ask why? This is because that we were assuming that the particles are distinguishable. However, it is not true when we are considering the mixing of two same gases! To elaborate the point, we may say that if we consider two microstates that differ from another merely in an interchange of two particles in different energy states, then according to our original mode of counting, we would regard these microstates as distinct.

Suppose there exist n_1 particles being in the state of ϵ_1, n_2 particles being in the state of ϵ_2, ..., and $n1 + n2 + ... = N$. The total number of permutations that can be effected among N particles distributed according to the set $\{n_i\}$

$$\frac{N!}{n_1!n_2!...} \tag{4.14}$$

This is correct for situations in which all N particles happen to be in different energy states but is certainly wrong for other situations. Why?

The recipe of Gibbs corrects the enumeration of the microstates only in a gross manner. Numerically, this would approach closer and closer to reality as the probability of n_i being greater. This in turn happens when the system at high temperature (so that many energy states are accessible) and has low density. However, quantum statistics will require better treatment.